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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 1	13 March 2023
Time : 1.5 hours	Maximum marks: 52
	Full marks: 50

- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
9	8	4	10	3	8	7	3	52

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Question 1. [9 points] Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 5 \\ 3 & 0 & 1 \end{pmatrix}$.

(a) Write A^{-1} as a product of elementary matrices. (Gauss reduce and show your working.)

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- (b) Write A as a product of elementary matrices. (Specify the entries of each elementary matrix explicitly.)

- (c) Suppose B is a square matrix and $B = E_k \dots E_2 E_1 U$ for some elementary matrices $E_k, \dots, E_2, E_1$ and U is in row echelon form. Does this imply that B is invertible? Justify your answer.

Question 2. [8 points] Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

- (a) Find all vectors $\mathbf{b}$ for which the equation $A\mathbf{x} = \mathbf{b}$ has:
- (i) infinitely many solutions
 - (ii) no solutions
 - (iii) a unique solution.

Show your working. [**Hint:** Write $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ and Gauss reduce the associated augmented matrix.]

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(b) Is A singular or non-singular? Give a reason for your answer.

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Question 3. [4 points] The following is an augmented matrix of a linear system, in the variables stated. Solve the system, giving your answer **in vector form**.

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 24 \\ 0 & 0 & 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad v, w, x, y, z.$$

Question 4. [10 points] Let

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 2 & 0 & 0 \\ -7 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

$$A_4 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_5 = \begin{pmatrix} 4 & 5 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 2 \end{pmatrix}, \quad A_6 = \begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

In each of (a) to (g) below, a description of a matrix is given. In each case, choose **one** matrix from the list above satisfying the description given. Write **only one of** $A_1, A_2, A_3, A_4, A_5, A_6$ in the appropriate block on the answer sheet. If it is impossible to find a matrix satisfying the description given, write “Impossible”. You may use a matrix more than once, or not at all.

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(a) A matrix that is not row-equivalent to I_3 .

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(b) A symmetric matrix in row echelon form.

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(c) A singular elementary matrix.

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(d) An upper triangular matrix which is invertible.

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(e) An elementary matrix that is not in row echelon form.

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(f) A matrix which can be written as the product of two elementary matrices, neither of which is an identity matrix.

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Also give the two elementary matrices, in the correct order.

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(g) A matrix which is not elementary and not in row echelon form, but can be written as the product of an elementary matrix and a matrix in row echelon form.

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Give the two matrices in the product, in the correct order.

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Question 5. [3 points] Suppose the $n \times n$ matrix A can be written as $A = E_1 E_2 \dots E_k$, where $E_1, E_2, \dots, E_k$ are elementary matrices. How can we write $2A$ as a product of elementary matrices?

Question 6. [8 points] Read the following proof and then answer the questions following it:

Proof: Let the $n \times n$ matrix A be row equivalent to I_n (1)
 Then there are elementary matrices $E_1, E_2, \dots, E_k$ such that $E_k \dots E_2 E_1 A = I$ (2)
 Hence $A = E_1^{-1} E_2^{-1} \dots E_k^{-1}$, (3)
 and therefore A is invertible. From this it follows that A is non-singular. □

(a) State the result proved here.

(b) Use the definition of row equivalence to explain how (2) follows from (1).

(c) Why does E_1^{-1} exist?

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(d) How does it follow from (3) that A is invertible?

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(e) Use only the definitions of invertibility and non-singularity to prove that an invertible matrix is non-singular.

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Question 7. [7 points] Consider the following statement:

For every $n \times n$ matrix A , if $A^{2023} = \mathbf{0}$, then $I_n - A$ is invertible.

(a) Write down the converse of the statement.

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(b) Write down the contrapositive of the statement.

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(c) Prove or disprove the statement.

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(d) Prove or disprove the converse of the statement.

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Question 8. [3 points] Suppose that A , and B are square matrices and $AC = CB$ for some invertible matrix C . If $B^k = I$ for some positive integer k , is it true that $A^k = I$?